

Muhammad Haris Ali

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Professional Profile

AI Engineer with hands-on expertise in Computer Vision, Generative AI, and Machine Learning. Experienced in LLM fine-tuning, edge AI deployment, and building intelligent systems that solve real-world problems. Proficient in Python, C++, and modern ML frameworks. Passionate about applying cutting-edge AI to healthcare, security, and education — with a track record of award-winning university projects and production-grade deployments.

Education

- Bahria University** Islamabad, Pakistan
Bachelor of Science in Computer Science 2021 – 2025

Professional Experience

- Government of Pakistan** Islamabad, Pakistan
AI Engineer (BPS-17) Sep 2025 – Present
 - Developing and deploying AI-driven solutions for national-scale digital transformation initiatives within a secure government environment.
 - Designing Computer Vision, Machine Learning, Natural Language Processing (NLP), and Large Language Model (LLM) based applications for mission-critical use cases.
 - Working on confidential government-level AI projects involving intelligent automation, document intelligence, decision support, and secure AI systems.
 - Building scalable AI services, REST APIs, and microservices for production deployment while ensuring security, reliability, and maintainability.
 - Optimizing deep learning models using TensorRT, ONNX Runtime, CUDA, and GPU acceleration techniques to achieve high-performance inference.
 - Setting up and maintaining local, on-premises GPU infrastructure for AI model deployment, including NVIDIA driver and CUDA Toolkit installation, system-level environment configuration, and GPU health monitoring.
 - Containerizing AI workloads with Docker and orchestrating multi-service deployments using Kubernetes for scalable, on-premises inference pipelines.
 - Configuring NVIDIA Multi-Process Service (MPS) and GPU partitioning strategies to enable efficient CUDA workload sharing and concurrent model serving on shared hardware.
 - Collaborating with multidisciplinary engineering teams throughout the complete AI lifecycle including data engineering, model development, deployment, monitoring, and continuous optimization.
- Techostudios** Remote
Generative AI Engineer | Financial AI Product Dec 2024 – Sep 2025
 - Architected an end-to-end AI platform for financial forecasting and transaction analysis using deep learning and Generative AI.
 - Developed LSTM-based forecasting models for equity trend prediction and anomaly detection across large-scale financial datasets.
 - Engineered time-series feature pipelines, including rolling statistics, technical indicators, and sequence windowing, to improve model signal quality and reduce noise in volatile market data.
 - Integrated Large Language Models to generate automated financial insights and analytical reports.
 - Designed prompt templates and retrieval workflows to ground LLM-generated summaries in verified transaction and market data, improving factual consistency of automated reports.
 - Managed the complete ML lifecycle including data preprocessing, feature engineering, model training, evaluation, deployment, and API integration.
 - Containerized model training and inference services with Docker to standardize environments across development and production.
- Freelance** Remote
Generative AI Engineer May 2023 – Present
 - Developed AI-powered interview assistants with speech recognition, LLM reasoning, and automated performance evaluation.
 - Built synthetic data generation pipelines using Large Language Models for healthcare and enterprise NLP applications.
 - Applied quality-filtering and diversity-sampling heuristics to synthetic datasets to reduce bias and improve downstream model generalization.
 - Designed multimodal semantic search systems using CLIP embeddings, vector databases, and Retrieval-Augmented Generation (RAG).
 - Developed AI-driven resume screening systems using transformer-based architectures for intelligent candidate ranking.
 - Built real-time Computer Vision solutions using YOLO for edge deployment on NVIDIA Jetson platforms.

- Optimized edge inference pipelines with model quantization and TensorRT conversion to reduce latency and memory footprint on resource-constrained devices.
- Designed enterprise Retrieval-Augmented Generation (RAG) systems integrating vector databases, document processing, and modern LLM orchestration frameworks.
- Set up local development and deployment environments using Docker and CUDA-enabled containers to ensure reproducible GPU-accelerated workflows across client projects.

• **RISETech, NUST College of E&ME**

Rawalpindi, Pakistan

• *Machine Learning Intern*

Jul 2024 – Aug 2024

- Fine-tuned Large Language Models including LLaMA and Gemma for healthcare and educational applications.
- Applied parameter-efficient fine-tuning techniques to adapt pretrained LLMs to clinical dialogue tasks with limited computational resources.
- Developed an AI-powered Medical OSCE simulation system supporting history taking, diagnosis, and automated performance assessment.
- Implemented custom datasets, prompt engineering strategies, and evaluation pipelines to improve model performance for clinical reasoning tasks.
- Collaborated with faculty supervisors to validate model outputs against clinical accuracy benchmarks and iterated on training data quality.

Projects

• **VetAI Chatbot (Final Year Project)**

• *Multi-Module Generative AI | Flutter | LLM Fine-Tuning | Voice AI*

- Built a multi-module AI chatbot for virtual veterinary services with a Flutter frontend and three integrated AI engines.
- Fine-tuned LLaMA for accurate disease diagnosis and treatment suggestions in veterinary contexts.
- Integrated Whisper for Urdu voice communication, improving rural accessibility.
- Used Gemini for conversational flow management and structured medical report generation.

• **ANPR Gate Entry System (Jetson Nano Edge AI Deployment)**

• *Computer Vision | YOLOv8 | Edge Processing | 2nd Position – Project Gala 2023*

- Deployed an Automatic Number Plate Recognition system on Jetson Nano for real-time edge inference.
- Trained custom YOLOv8 model on self-curated Pakistani number plate dataset with OCR post-processing.
- Achieved low-latency on-device inference with no cloud dependency.
- Awarded 2nd position at Project Gala 2023 for innovation in embedded AI.

• **HACRF Spectrum Analyzer for Drone Detection**

• *Edge AI | Signal Processing | ARM/x64/AMD | Confidential Client*

- Developed a spectrum analysis solution for drone detection, processing time-varying RF frequency data.
- Built hardware-specific implementations targeting ARM, x64, and AMD architectures for edge processing.
- Implemented algorithms to detect characteristic frequency signatures of drones with low false-positive rates.
- Designed for autonomous operation in contested environments with minimal latency.

• **Synthetic Data Generation Pipeline for Healthcare NLP**

• *LLM | Data Augmentation | Healthcare AI*

- Created a pipeline using LLMs to generate 50K+ synthetic labeled text samples for healthcare NLP tasks, reducing manual annotation costs by 80%.
- Implemented quality filtering and diversity sampling to ensure dataset quality and coverage.
- Used generated data to improve downstream classifier accuracy by 15% over baseline.

• **Cancer Cell Segmentation & Classification**

• *Computer Vision | U-Net | Medical Imaging | Deep Learning*

- Developed a deep learning pipeline for automated cancer cell segmentation using U-Net architecture.
- Benchmarked against ResNet and EfficientNet classifiers for cell-type identification.
- Applied data augmentation, transfer learning, and class imbalance handling techniques.

• **SAM 3D: Sketch-to-Animation for Toddlers**

• *Computer Vision | Segment Anything Model | 3D Rendering | EdTech*

- Built an interactive learning tool converting children's sketches into animated 3D characters.
- Used Meta's Segment Anything Model for precise sketch segmentation.
- Implemented sketch-to-3D pipeline with model rigging and procedural animation.

• **RAG System for Enterprise Knowledge Bases**

• *LLM | Retrieval-Augmented Generation | Vector Search*

- Developed a RAG pipeline for enterprise document retrieval and Q&A, integrating ingestion, chunking, vector embeddings, and LLM orchestration.
- Deployed using LangChain, OpenAI embeddings, and Pinecone vector database with sub-second query latency.
- Tuned chunking strategy and retrieval top-k parameters to balance answer relevance against response latency for enterprise-scale document sets.

• Document Summarization Tool

- *NLP | TF-IDF | PyMuPDF | Flask Web App*
 - Developed web application for PDF text extraction and summarization using TF-IDF scoring with NLTK.
 - Built a Flask-based interface allowing users to upload PDFs and receive extractive summaries with adjustable compression ratio.

• Real-Time Inventory Detection with YOLOv8 on Jetson

- *Computer Vision | YOLOv8 | Edge AI | Retail Automation*
 - Built an edge-optimized inventory management system using YOLOv8 deployed on NVIDIA Jetson Nano for real-time object detection and stock monitoring.
 - Optimized model inference with TensorRT conversion and batch processing to maintain real-time frame rates on constrained edge hardware.

Skills

- **AI & Deep Learning:** LLM Fine-Tuning, Computer Vision, LSTM, U-Net, YOLOv8, SAM, Generative AI, NLP, Time-Series Forecasting, Signal Processing, Transformer Architectures, Attention Mechanisms, Retrieval-Augmented Generation (RAG), Stable Diffusion, CLIP, LoRA, Embeddings
- **ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, spaCy, NLTK, OpenCV, FastAI, XGBoost,
- **Model Optimization:** TensorRT, ONNX, OpenVINO, CUDA, CUDA Slicing, MPS, Model Pruning, Quantization, Knowledge Distillation, GPU Acceleration
- **DevOps & MLOps:** Docker, Kubernetes, NVIDIA Container Toolkit, AWS (EC2, S3), GCP, MinIO, FastAPI, Flask, REST APIs, Microservices
- **Local AI Deployment & GPU Systems:** NVIDIA CUDA Toolkit Setup, NVIDIA Drivers, CUDA MPS (Multi-Process Service), GPU Partitioning, nvidia-smi, On-Premises GPU Infrastructure Configuration
- **Programming Language:** Python, C++, PHP, Laravel, Bash
- **Databases:** SQL, PostgreSQL, MySQL, MongoDB, Redis, FAISS
- **Platforms:** Linux, Git, GitHub Actions, MLflow, Kubeflow, Jupyter, NVIDIA Jetson Nano, Raspberry Pi

Certifications

- **Google Data Analytics Professional Certificate** – Google
- **AWS Academy Cloud Architecting** – Amazon Web Services
- **Introduction to Data Science** – IBM
- **AI for Everyone** – Google
- **Agentic AI – Hands-On Training** – Agentic workflows, autonomous AI systems, and tool-use patterns
- **Deep Learning Specialization** – Coursera (In Progress)

Core Competencies

- **Technical Leadership:** Owning AI systems end-to-end from research and prototyping through production deployment and monitoring.
- **Cross-Functional Collaboration:** Working with data engineers, backend teams, and stakeholders to translate requirements into deployable AI solutions.
- **Problem Solving:** Diagnosing performance bottlenecks in training and inference pipelines and applying targeted optimization strategies.
- **Communication:** Translating technical AI concepts into clear insights and recommendations for non-technical stakeholders.

Achievements

- **2nd Position – AI, Cyber & Drone Swarm Gala** – Awarded for a proof-of-concept facial image enhancement system that processes CCTV footage of drivers and co-passengers inside vehicles to improve image quality for downstream facial recognition.
- **3rd Position – Project Gala 2023**, Bahria University – Awarded for the ANPR Gate Entry System, an embedded edge-AI solution deployed on NVIDIA Jetson Nano.

Languages

- **English:** Professional Working Proficiency
- **Urdu:** Native